



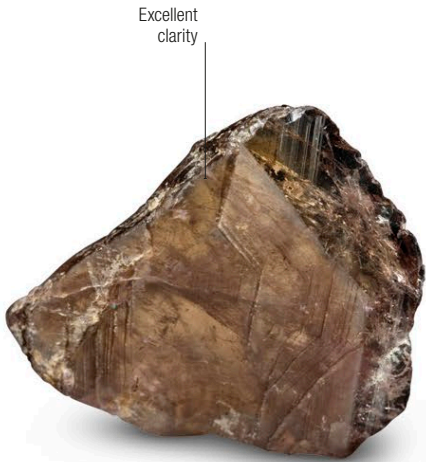
Axinite

△ Emerald-cut axinite from Mexico

Axinite refers to a group of four minerals, which in rough form are virtually indistinguishable and are structurally identical. The name derives from the Greek *axine*, meaning “axe”, which refers to the sharp, hard crystals. The most common colour is clove-brown; varieties can also be grey to bluish-grey, honey-, grey-, or golden-brown; pink, violet-blue, yellow, orange, or red. Its gemstones are easily chipped, so are usually only faceted for collectors. Axinite is piezoelectric and pyroelectric, meaning it generates electricity when stressed, or rapidly heated or cooled, respectively.

Specification

Chemical name Calcium-, iron-, manganese- aluminium borosilicate | **Formula** $(Ca_2Fe, Mn, Al_2)(BSi_4O_{15})(OH)$ | **Colours** Various | **Structure** Triclinic | **Hardness** 6.5–7 | **SG** 3.2–3.3 | **RI** 1.67–1.70 | **Lustre** Vitreous | **Streak** Colourless to light brown | **Locations** USA, Russia, Australia, Mexico, France, Sri Lanka



Axinite rough | Rough | This facet-grade piece of axinite rough has good colour and clarity, and retains some of its original crystal form. The tabular structure is typical.



Axinite in matrix | Rough | This rock matrix holds a number of red-brown axinite crystals showing classic axe-head crystal forms.



Axinite crystals | Rough | These two crystals, each of which exhibits classic and perfect axinite form, have intergrown at one of the facial angles.



Oval cushion | Cut | Although there are some natural inclusions in one end of this step-cut oval cushion, it is still a desirable gem.



Blue oval | Colour variety | This oval brilliant is unusually fine both in clarity and colour: most axinite is golden- or reddish-brown, so this hue is relatively uncommon.



Mexican axinite | Colour variety | This cushion-cut axinite gem displays a fairly typical reddish-orange colour. It weighs 4.29 carats and has good transparency.

Prismatic crystal | Rough | Although axinite is typically found as thin, hard axe-head-shaped crystals, it can also occur in blocky form, as in this specimen.

